Organisational closure in dynamical systems

Sept 3rd, 2004

Boris.saulnier@ens.fr

« Organisational closure »?

Concepts and theories :

- Autopoiesis (Maturana,Varela)
- Closure under efficient causes (Rashevsky, Rosen)
- Individuation (Simondon, Thom)
- Auto-organization (Atlan)
- Dissipative structures? (Prigogine)
- Integration? (Edelman, Sporns, Tononi)

- Simply means : differents parts are organized in a whole, an can not exist without the whole

- But we should not speak of organisational cloture for the shell of an egg…

- Intuitions : dynamics coupling, viability, control theory, multistationarity and circuits analysis, synchronisation and chaos control…

Mysterious system…

- You are given the <u>local</u> dynamics :

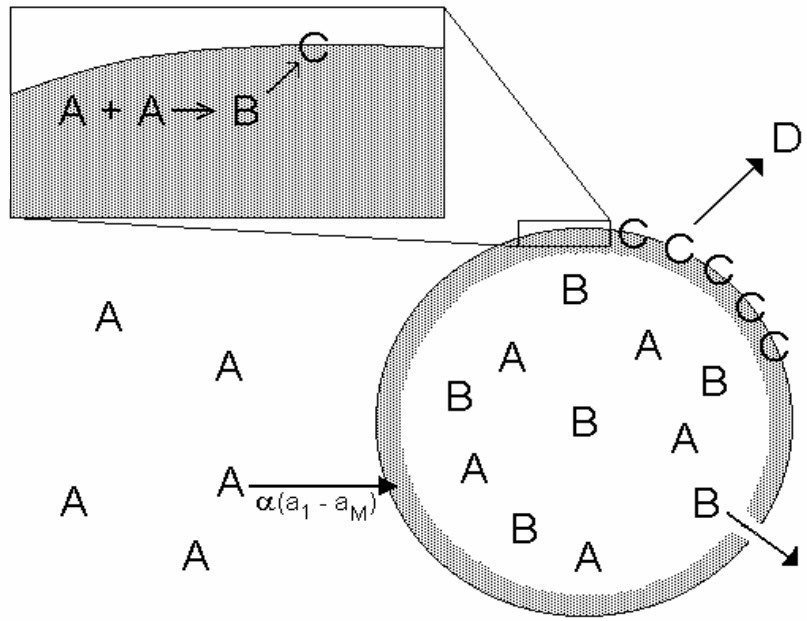
$$\begin{cases} dc_M/dt = -k_c \,.c_M + k_b \,.b_M \,.(c_1 - c_M) & (1) \\ \partial a/\partial t = \Delta a - 2\, k_a \,.a^2{}_M \,.c_M \,.(b_1 - b_M) \,.\, \delta_M & (2) \\ \partial b/\partial t = \Delta b + k_a \,.a^2{}_M \,.c_M \,.(b_1 - b_M) \,.\, \delta_M & (3) \end{cases}$$

- And you know :
  - Constants : Kc, Kb, C1, B1
  - Functions of time : Cm, Bm, Am
  - Functions of time and space : a, b
  - δm is a characteristic function for a given zone in space
- Can you see closure somewhere?
  - It seems very difficult to give an interpretation
  - From 1 : Cm evolves in time, not in space. Created at a rate proportional to Bm and (C1-Cm). Vanishes at rate proportional to Cm.
  - From 2 and 3 :a and b evolve in time and space. Laplacian represents a diffusion term. a vanishes at a rate which is twice b creation rate. This creation rate depends on Am, Cm and Bm.

And the winner is…the tesselation automata!

- Actually the system has a <u>membrane</u>
- δm is the characteristic function for a thin layer just underneath the membrane
  - The reaction A+A->B is catalyzed by the inside surface of the membrane (C catalyses the production of B)
- Bm, Cm and Am are concentrations for products in a close vicinity to the membrane
- The intact membrane is impermeable to the B components
  - a and b are concentrations of A and B in the volume closed by the membrane

$$\begin{cases} dc_M/dt = -k_c \,.c_M + k_b \,.b_M \,.(c_1 - c_M) \\ \partial a/\partial t = \Delta a - 2\, k_a \,.a^2_M \,.c_M \,.(b_1 - b_M) \,.\, \delta_M \\ \partial b/\partial t = \Delta b + k_a \,.a^2_M \,.c_M \,.(b_1 - b_M) \,.\, \delta_M \end{cases}$$



Is it enough to know the local dynamics?

- The knowledge of the spatial organisation is necessary to interpret and integrate the equations
- We should be able to derive the existence of the membrane from the local dynamics equations
- A vicious circle?
    - The membrane existence depends on the local dynamics
    - The local dynamics depends on the existence of the membrane
- Plus : the « phase space change » problem
  - If the membrane disrupts the local dynamics equations are not valid anymore.
  - There is a validity domain for these equations, depending on the values of the variables (here if Cm is too low it means there is no more membrane)
  - When evolving, the dynamics can change

Dynamic analysis of the model

- The key variable is C*=C1/cm
  - (C1:max concentration of Cm=membrane with no holes;Cm=concentration of C in the membrane)
- Percolation analysis shows critical value Cc* : below Cc* the membrane separates into different pieces
- A single fixed-point attractor PF1
  - The value of C* in PF1 depends on a number of parameters
  - Can vary from 99% to 1% without discontinuity : not satisfying
  - Need to modify the model

Modified model

- In model I, the membrane was unconditionnaly impermeable to the escape of B
- Model II (modified): $dc_M/dt = - k_c \,.c_M + k_b \,.b_M \,.(c_1 - c_M).\, p_r(c_M).$
  - (I'd rather modify equation 3…)
- Result :
  - PF1 (stable) like in model1, exept that C*>50% (necessarily)
  - Plus : PF0 (stable) where C*=0 (collapsed system)
  - Plus PF2 : bifurcation point. Unstable

(1bis)

- Qualitatively :
  - Above PF2 : disintegration of the membrane balanced by the repair process, c*>50%, fragmentation of the membrane can not occur. System may waver, but then recover (structural stability)
    - » ALIVE
  - Below PF2 : positive feedback, more and more holes, less and less catalysation, collapse of the system
    - » DEAD

From the dynamic systems point of view :

- Phase space problem :
  - There is still a bigger and UNIQUE phase space, where to represent the evolution of the system
- In this big phase space, being alive means :
  - A given subset of the variables respect constraints, ie stay in a given volume of the phase space
  - The system resist perturbations : does not quit its attraction basin even for (extrinsic) variations of environment variables
- Is the dynamic system really specific?
  - Are autopoietic or cognitive capabilities IN the dynamic system description?
  - Are they relative to the observer « reading » of the equations?
    - Separation into variables and parameters
    - Separation into system Vs environment
    - Separation between components and « network of processes »
    - Separation between « inputs » and « outputs »
- The maths see only : trajectories, flows, basins, attractors, bifurcation points…
  - « Systems are not in Nature, they are in the mind of humans » (Claude Bernard)

Definitions (Stewart, Bourgine 2004)

- Autopoietic system :
  - a network of processes that produces the components that reproduce the network
  - and that also regulates the boundary conditions necessary for its ongoing existence as a network
- Cognitive system : sensory inputs serve to trigger actions in a specific way, so as to satisfy a viability constraint
- Living = autopoietic+cognitive

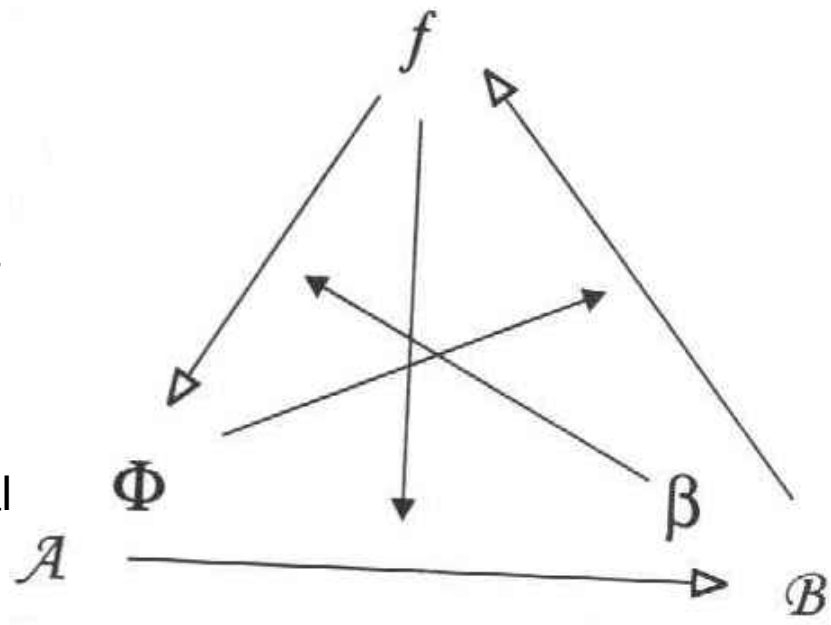
« network of processes that produce components that… » : an interpretation

- Rosen interpretation of Aristotle causes:
  - Consider the <u>relation</u> : f:(A->B)
  - A and B are sets; f is a map
  - A is the material cause for B ; f is the efficient cause for B
  - f applied to A is the formal cause for B
- Physics
  - The science of SDDS (State Determined Dynamical Systems) : dX/dt=f(X)
  - f : (X(t) -> X(t+dt))
  - States are material causes ; the dynamics f is the efficient cause ; integration is the formal cause
- (My)interpretation of Rosen interpretation :
  - See f as a process
  - See A and B as components

Biology : functions and final causes

- Needs « functions », and final causes
- For Rosen, living = closed under efficient causality
- When asking « why f ? », you want to find an answer INSIDE the system
- Gives the following minimal configuration :
- Why f?
    - Because Φ (efficient cause)
    - Because B (material cause)
    - Because f is the (material) cause for Φ (final cause)
    - Because f is the (efficient) cause for Bβ (final cause)

- Note that the final cause is not a component nor a process. It is a property :
  - As a component :being the material cause of your efficient cause
  - As a process: being the efficient cause of your material cause
- Consequence : components (and processes) have a (biological) function
- For Stewart (2002) : SDDS and and relational models are not commensurables ; but WHY? Can we actually conciliate them?



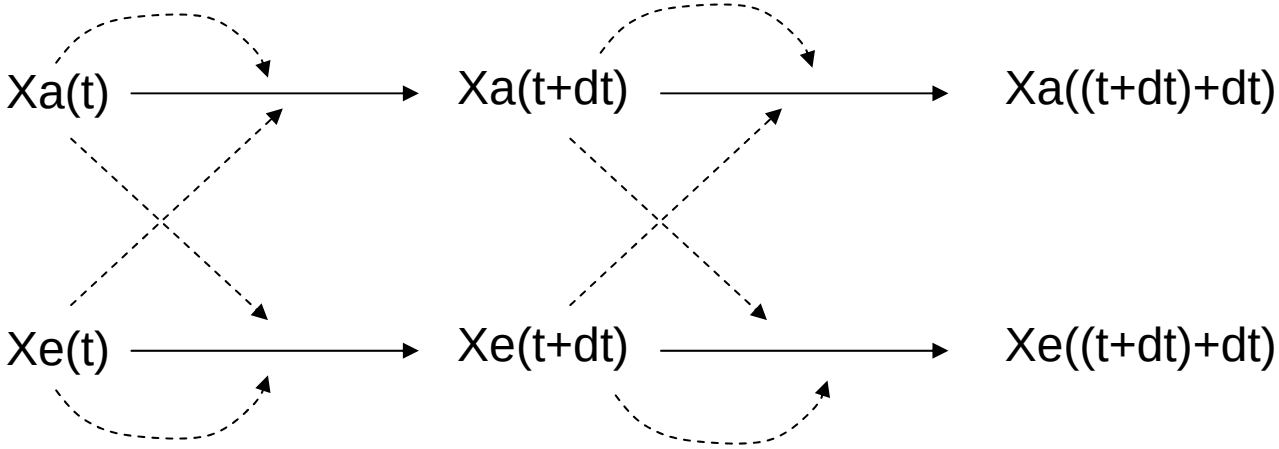
Trying to conciliate dynamical systems and relational models

- Remember :
  - f is the efficient cause for B : f is a process
  - But f is also a component!
    - » It means the dynamics has a material cause…
    - » And this cause is an other dynamics…
  - How can you be state and dynamics at the same time??
- Autonomous Vs Non autonomous dyn. Systems
  - A dyn sys is « non autonomous » if its parameters are allowed to vary in time (Beer 1995)
  - Dx/dt=f(x;u(t))
  - « We can think of such parameters as <u>inputs</u> to the system »
  - These inputs can be considered as states of an other system
  - And they <u>contribute to specify</u> the dynamics of the system
  - The coupling of systems « make » a dynamics out of a state

Relation between states/parameters and dynamics in coupled dynamical systems

- Ashby's notation (Design for a brain):
  - dXa/dt=f(Xa,Ua) ; dXe/dt=g(Xe,Ue)
  - a=agent; e=environment; x=states; u=parameters
  - Ua=S(Xe) : sensory function
  - Ue=M(Xa) : motor function
  - Some parameters are functions of the state variables of the other syetm
  - For Ashby variables are associated to changes of state, and parameters are associated to « a change from one behaviour to the other »
- Beer's notation:
  - dXa/dt=f(Xa,S(Xe),U'a) ; dXe/dt=g(Xe,M(Xa,U'e)
  - U'a and U'e are parameters that do not participate in the coupling

Suggests a graph vision of interactions



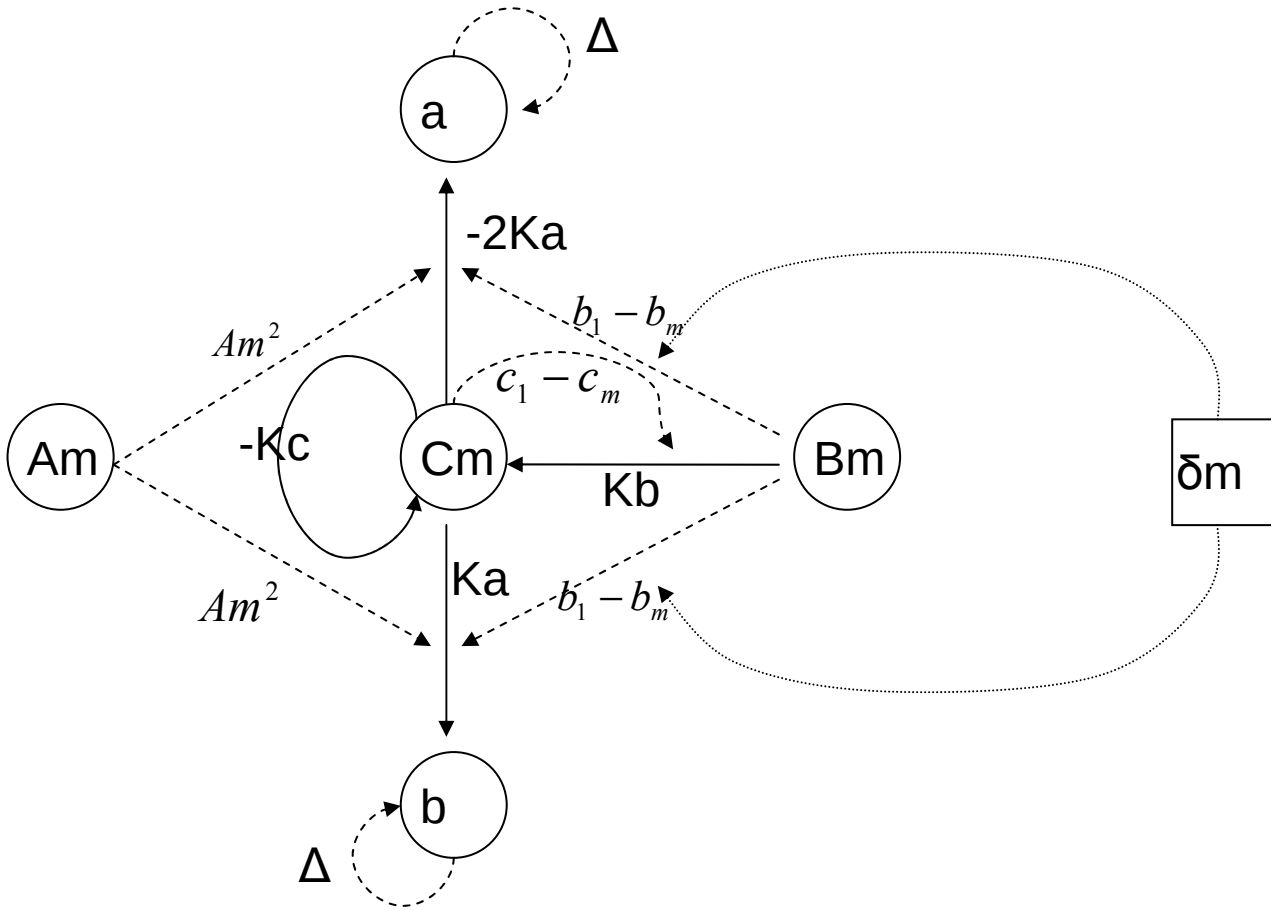
Material cause :

Efficient cause :

Why Xa(t+dt)?

- Because Xa(t) : material cause
- Because Xe(t) and f : efficient cause
- Because is the cause of Xa(t+dt+dt) (final cause = perpetuation)
- Because is the cause of Xe(t+dt+dt) (perpetuation)

Tesselation automata



One can see :

- The central position of Cm and its high connectivity
- The « heteronomic » aspect of δm
- The feedback circuits
- Symetries

$$\begin{cases} dc_M/dt = -k_c \,.c_M + k_b \,.b_M \,.(c_1 - c_M) \\ \partial a/\partial t = \Delta a - 2\, k_a \,.a^2{}_M \,.c_M \,.(b_1 - b_M) \,.\, \delta_M \\ \partial b/\partial t = \Delta b + k_a \,.a^2{}_M \,.c_M \,.(b_1 - b_M) \,.\, \delta_M \end{cases}$$

Insights about the coupled dynamics

- The traditional concepts (attractors, basins of attraction etc.) apply only on timescales small relative to the timescale of the parameter variation
- Because of the feedback between the two systems, each of the two is continuously deforming the flow of the other
  - Drastic deformation of the flow if any coupling parameter crosses bifurcation points
- A dynamical system follows a trajectory specified by its own current state and dynamic laws
  - The second system can bias the intrinsic « tendencies »
  - Rather than one seeing one system steering the other, we may beter see each as a perturbation for the other
- Coupled nonautonomous systems A and E can be viewed as a <u>single</u> autonomous dynamical system
  - In this case, as previously, the separation into two systems, states and parameters, inputs and outputs seems completely relative to the observer reading : subjective aspect of these notions

Decomposing a single autonomous dynamical system in coupled nonautonomous systems

- Consideration of different timescales
  - Parameters on longer timescales than states
  - Timescales may denote organisational levels
- Autonomous subsets of state variables
  - Ie : not dependent of other variable variation = invariance of the global flow by translation of any vector in the considered subspace
- Defining function
  - Let's consider the interaction of systems A and B
  - The resulting dynamic for A is d
  - Let's call function Fa,b of A in its interaction with B the geometric structure of the phase portrait of d
  - We might have Fa,b~Fc,b
    - » Functional degeneracy : different structures realize the same function
  - We might have Fa,b¬~Fa,d
    - » Structural degeneracy : the same structure realize different functions in different environments
- Modularity and compositionality
  - Module : a subset of variables constituing a system with a limited number of outputs and outputs.
  - A module can be commanded, ie you can select its dynamics from outside and « choose » among a limited number of patterns
  - « The cat is eating » Vs « the dog is eating » : compositionality comes from the fact that « the dog » and « the cat » have a common « near autonomous subsystem »…

Analysing the interaction of dynamics

I’m investigating different tools and approaches :

- Circuits and feedback analysis (Thomas)
- Viability theory (Aubin)
- Multiscale analysis (Lesne)
- Theory of control (Tyson/Novak)
- Coupled cell theory (Golubitsky)
- Coupled oscillators (Kuramoto/Winfree)

Networks as a representation of system interactions

- Allow to investigate :
  - structure and function relations
  - Understand function in dynamical terms
  - Local/global relations

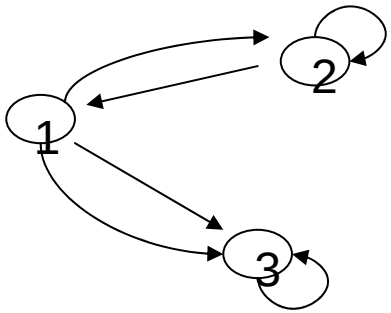
- More and more, artificial neural networks are « pretexts » to implement, model, analyse a dynamic with required characteristics

- Beyond the phenomenological conformity of the phenomenon and constructed dynamics…
  - Are networks in nature?
  - Do they speak of the reality?
  - Suggestion : structural realism…

Jacobian and graph vision of ODEs

$$\begin{pmatrix} 0 & 1 & -2x_1 \\ 2 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\begin{cases} \dfrac{dx_1}{dt} = x_2 - 2x_3x_1 \\ \dfrac{dx_2}{dt} = 2x_1 - x_2 \\ \dfrac{dx_3}{dt} = x_1 + x_3 \end{cases}$$



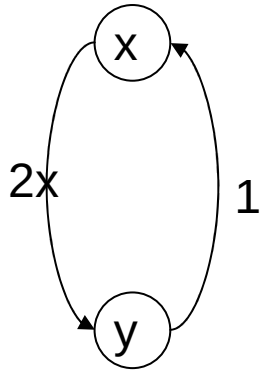
Example : 2 variables

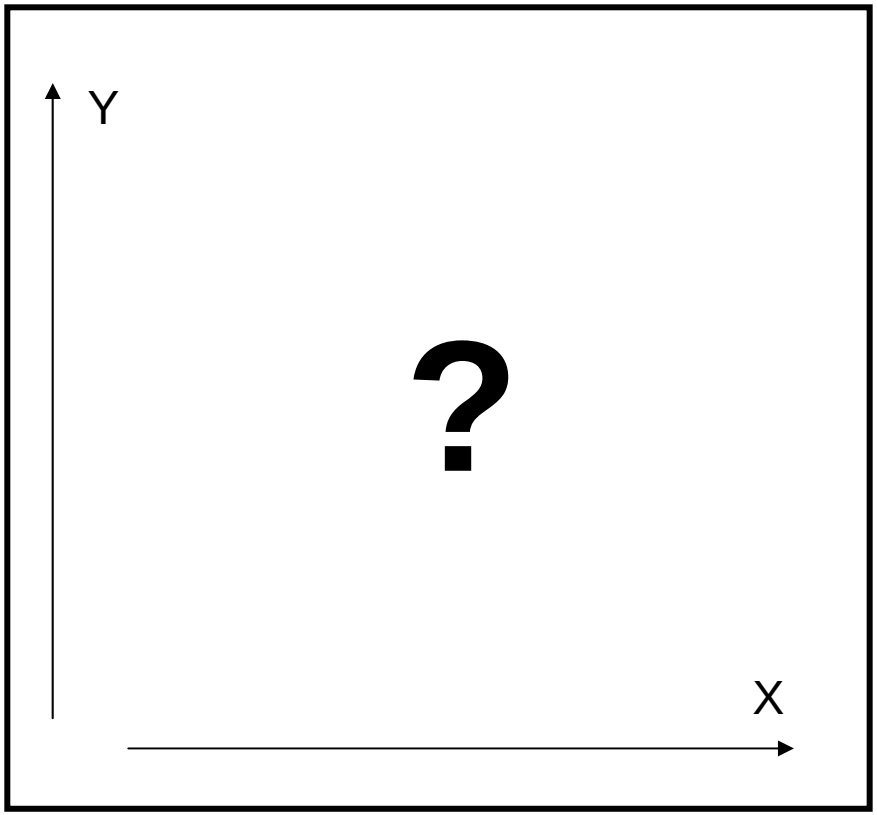
Pure full-circuits in 2 variables		
The 2-circuit	Eigenvalues	Steady state
Positive $\begin{pmatrix} & - \\ - & \end{pmatrix}$ or $\begin{pmatrix} & + \\ + & \end{pmatrix}$	$(+,-)$	Saddle point
Negative $\begin{pmatrix} & + \\ - & \end{pmatrix}$ or $\begin{pmatrix} & - \\ + & \end{pmatrix}$	complex	Center[a]
The unions of two 1-circuits		
$\begin{pmatrix} - & \\ & - \end{pmatrix}$	$(-,-)$	Stable ''node''
$\begin{pmatrix} - & \\ & + \end{pmatrix}$ or $\begin{pmatrix} + & \\ & - \end{pmatrix}$	$(+,-)$	''Saddle point''
$\begin{pmatrix} + & \\ & + \end{pmatrix}$	$(+,+)$	Unstable ''node''

Ambiguous circuit

$$\begin{cases} \dot{x} = y \\ \dot{y} = x^2 - 1 \end{cases}$$

$$M = \begin{pmatrix} 0 & 1 \\ 2x & 0 \end{pmatrix}$$



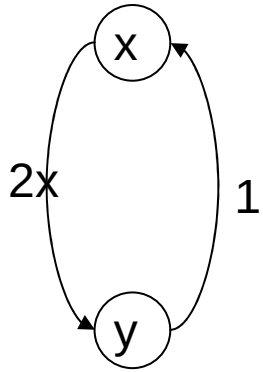


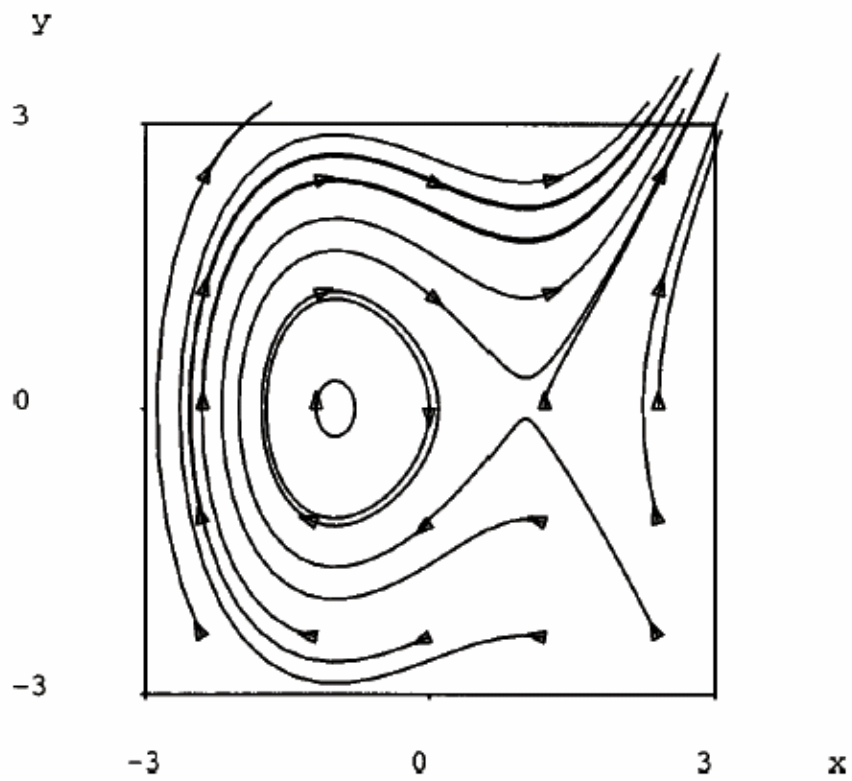
- Phase space made of 2 zones
- If $x > 0$ positive circuit (steady state = saddle point)
- If $x < 0$ negative circuit (steady state = center)
- 2 equilibria : (-1,0) and (+1,0)

Exemple de circuit ambigu

$$\begin{cases} \dot{x} = y \\ \dot{y} = x^2 - 1 \end{cases}$$

$$M = \begin{pmatrix} 0 & 1 \\ 2x & 0 \end{pmatrix}$$





- Phase space made of 2 zones
- If $x > 0$ positive circuit (steady state = saddle point)
- If $x < 0$ negative circuit (steady state = center)
- 2 equilibria : (-1,0) and (+1,0)

Systèmes dynamiques et boucles de rétroaction : 2 conjectures

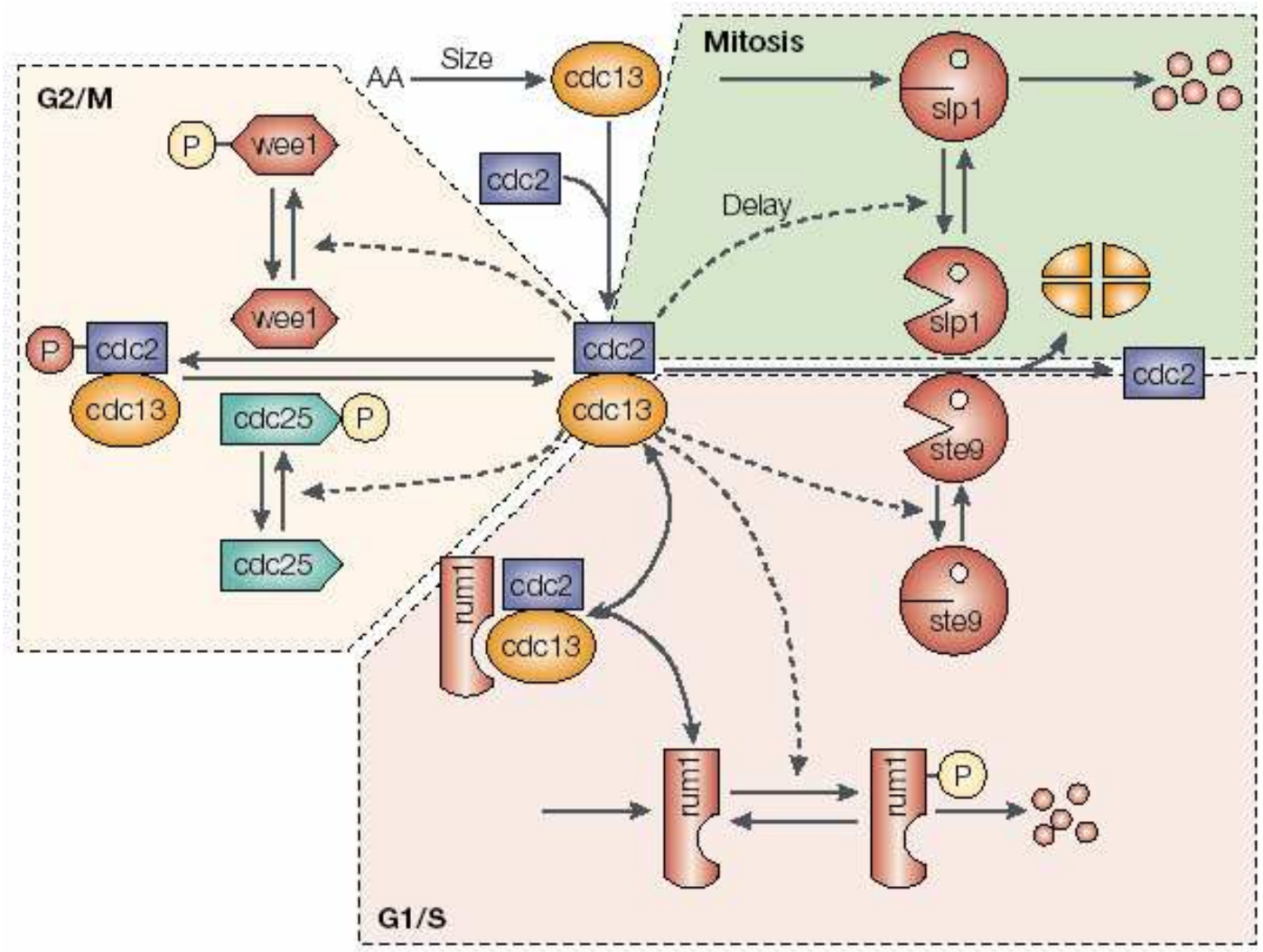
- La biologie inspire une analyse des dynamiques non linéaires en termes de boucles de rétroaction (positives ou négatives)

- Conjecture 1 (R. Thomas, 1981)
  - Une boucle de rétroaction positive dans le graphe d'interaction d'un système différentiel est une condition nécessaire de multistationnarité (existence de plusieurs équilibres)
  - Exemple en biologie : multistationnarité = différenciation cellulaire

- Conjecture 2 (R. Thomas, 1981)
  - Une boucle de rétroaction négative dans le graphe d'interaction est une condition nécessaire de comportement périodique stable
  - Périodicité stable très importante également en biologie

- Aspect étonnant : ces conjectures dépendent très peu de la formulation précise du système dynamique sous-jacent
  - Ne dépendent que de conditions sur les signes de la matrice jacobienne
  - Cad la façon dont une variable influence la variation temporelle d'une autre variable

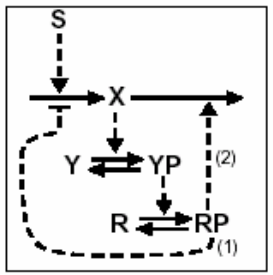
Réseau de contrôle de la division cellulaire de la levure de fission *Schizosaccharomyces pombe*

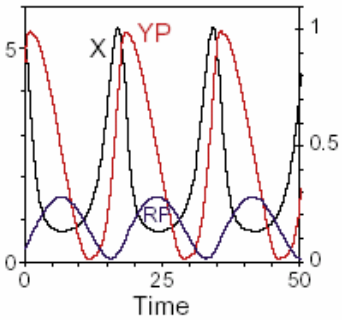


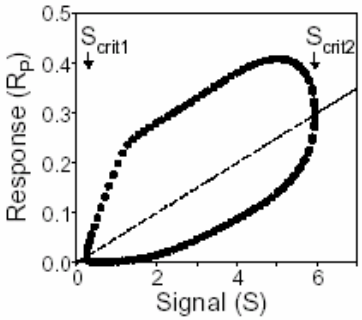
Système divisé en trois modules, qui régulent les transitions de la phase G1 à la phase S, et de G2 en M, ainsi que la transition vers la mitose.

Méthode

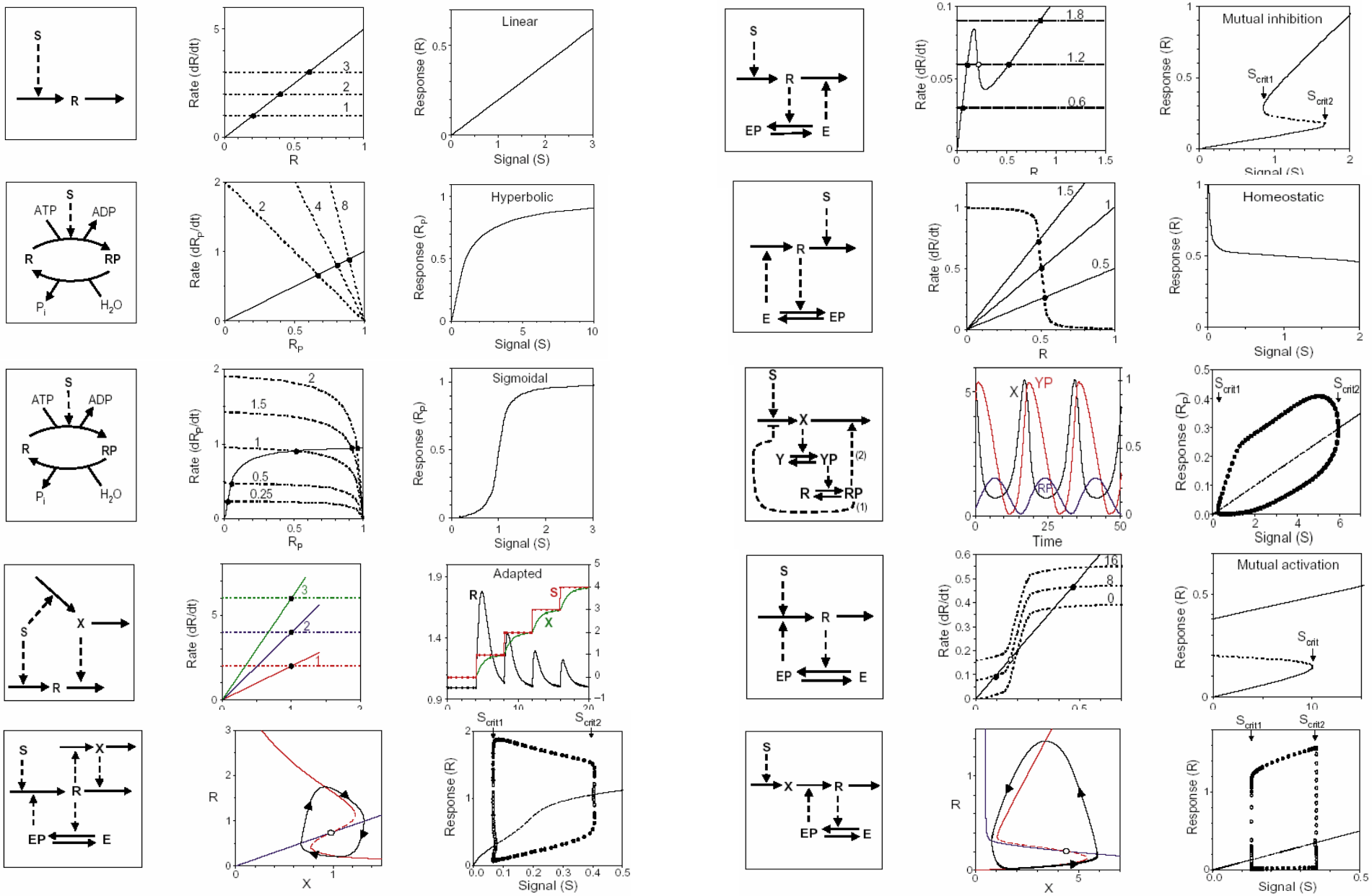
- On dessine un « diagramme de câblage biochimique »
- Chaque nœud représente une protéine spécifique
- Les flèches représentent les processus qui produisent (ou consomment) et activent (ou désactivent) la protéine
- En plus les protéines régulent les conversions biochimiques
- Le biologiste voudrait déduire/expliquer la physiologie cellulaire à partir de la dynamique du réseau
- Il propose un découpage en modules
- Ces modules sont analysés en fonction d'un nombre limité de « dynamiques de base »







Mathématisation de l'analyse locale à l'œuvre en bio moléculaire



What are functions of genes?

Signal transduction: sensing a physical signal and turning into a chemical signal

Structural support: creating the shape and pliability of a cell or set of cells

Enzymatic catalysis: accelerating chemical transformations otherwise too slow.

Transport: getting things into and out of separated compartments

Movement: contracting in order to pull things together or push things apart.

Transcription control: deciding when other genes should be turned ON/OFF

Trafficking: affecting where different elements end up inside the cell

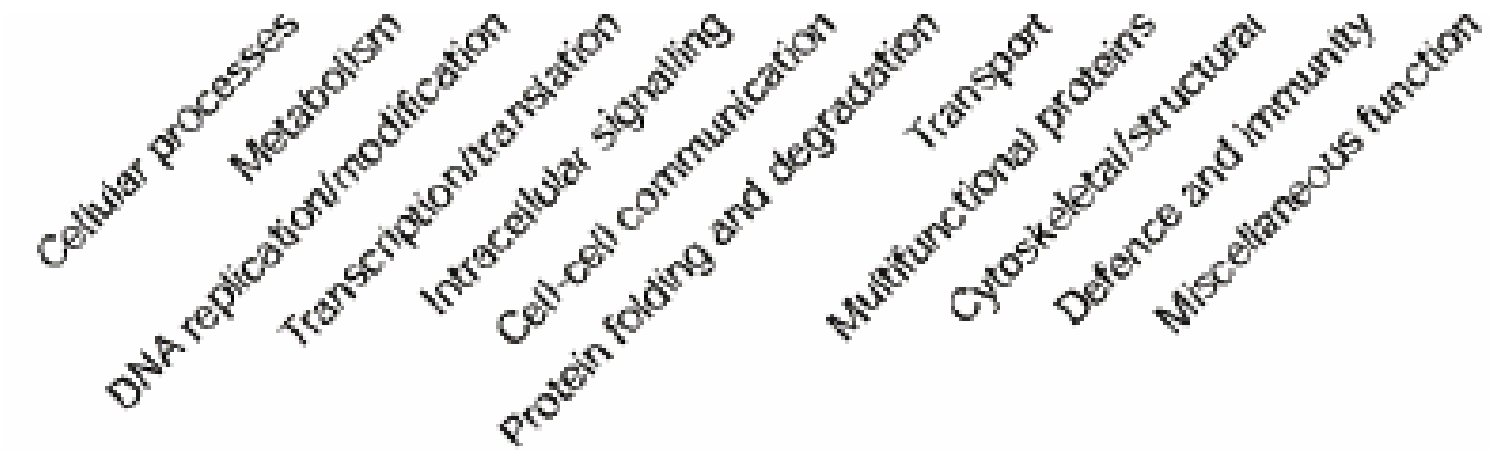


Figure 37 Functional categories in eukaryotic proteomes. The classification categories were derived from functional classification systems, including the top-level biological function category of the Gene Ontology project (GO; see http://www.geneontology.org).